

WebFX Documentation

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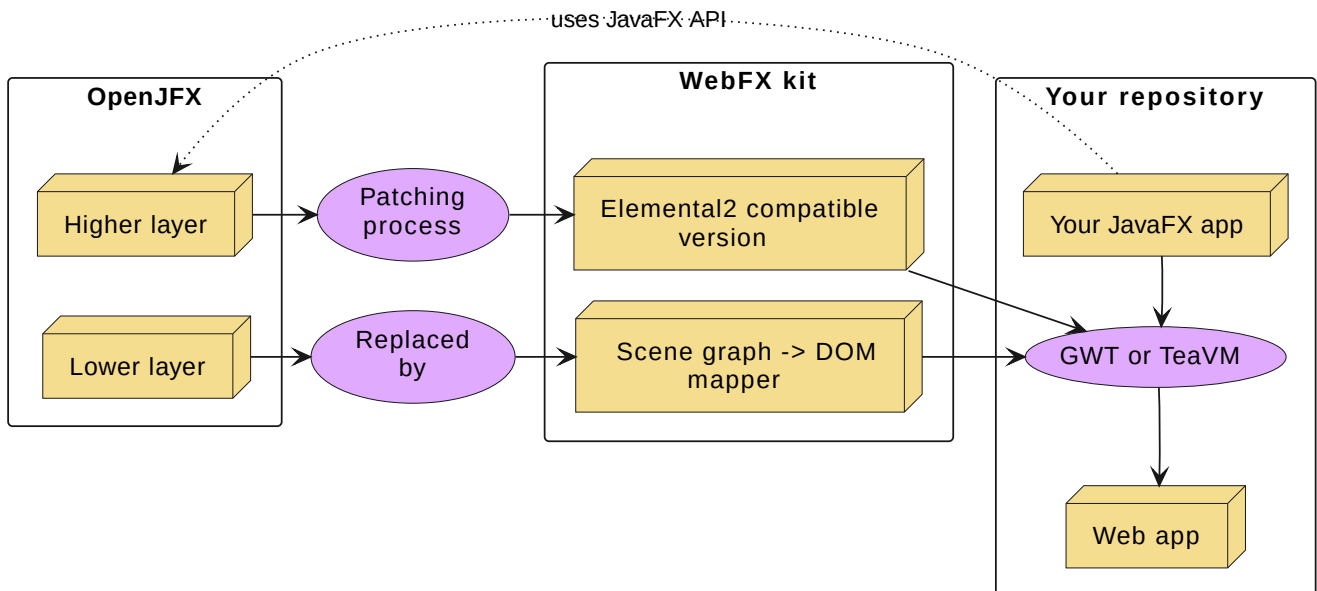
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Introduction

What is WebFX?

WebFX is a JavaFX application transpiler powered by [GWT](#) or [TeaVM](#). It can transpile a JavaFX application into a standalone, self-contained JavaScript or WebAssembly web app (with no plugin or server required for execution in the browser).

How it works



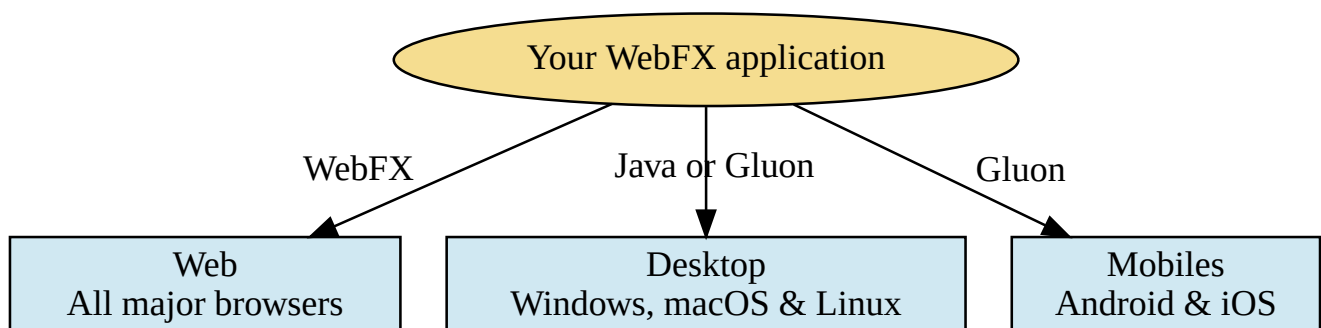
The [WebFX kit](#) is the heart of WebFX. It's a modified version of OpenJFX that can be transpiled by GWT and TeaVM. This is achieved by patching the higher layer of OpenJFX (which contains the main JavaFX features and API) to make it GWT and TeaVM compatible, and by replacing the lower layer (the graphic rendering pipeline) with a scene graph → DOM mapper (rendered by the browser).

Benefits

No server

There are already great solutions to run Swing or JavaFX applications in the browser without plugins by actually running them on a server. Those solutions avoid some current WebFX limitations, but a standard self-contained web packaging is a much simpler, more scalable, and more reliable execution model. This is precisely the benefit of WebFX, and probably the main reason why you would prefer it over the other existing solutions.

Cross-platform



In addition to the web platform, a WebFX application can also run on desktop and mobile. WebFX will invoke the standard Java toolchain to produce desktop executables (with an optimized JRE embedded), and the Gluon toolchain to produce native executables for desktop and mobile.

Some alternative technologies or JVM languages allow you to do the same but only for the

application logic, because they don't offer a cross-platform UI toolkit. With WebFX, you can do full cross-platform development of your entire application from a single codebase.

Java full-stack

Writing your whole stack in Java is a big advantage, keeping your environment simple and homogeneous from a single Java IDE. Here, full-stack means you can use Java on both the frontend and backend. This prevents you from having to master other complex ecosystems such as JavaScript or TypeScript, and also allows you to share common code between your backend and frontend using the Java module system - a great advantage compared to heterogeneous systems.

Performance

Despite the large size of OpenJFX, WebFX can produce lightweight web apps, as demonstrated by the demos:

WebFX demo	GWT JS size *	TeaVM JS size *	TeaVM Wasm size *
Colorful circles	97 kB	176 kB	220 kB
Tally counter	113 kB	200 kB	253 kB
DemoFX	147 kB	241 kB	303 kB
Modern gauge	157 kB	225 kB	278 kB
Mandelbrot	159 kB	247 kB	310 kB
SpaceFX	171 kB	259 kB	331 kB
FX2048	206 kB	341 kB	428 kB
Enzo clocks	296 kB	371 kB	484 kB

* compressed JS/Wasm size transferred over the network, excluding images and other resources

The secret? Three things:

- The scene graph → DOM mapper is much thinner than the original OpenJFX lower layer, which has to reimplement many features a browser already has.
- GWT and TeaVM run a dead code elimination (tree-shaking) process, which removes the JavaFX classes not used by the WebFX application.
- GWT produces very compact and optimized JS code, leading to highly performant web apps.

Gradual learning curve

WebFX is not an additional UI toolkit to learn - it is an emulation of the well-known and well-documented JavaFX API. All the powerful JavaFX features that you know and love are available for your web app. You will feel right at home with WebFX!

Fast development cycles

You don't need to run regular GWT/TeaVM compilations as in traditional GWT/TeaVM development,

because you can already run and debug your WebFX application directly in your Java IDE with the OpenJFX runtime. You typically transpile your app only at the end of a development cycle to check the web output, after you have finished developing a feature using the standard JavaFX development model.

Free and open source

WebFX is an open source initiative released under Apache 2.0 license.

Limitations

The WebFX kit coverage is currently limited to the essential features of JavaFX. So to successfully compile to the web, your JavaFX code needs to meet these two requirements:

- use only the features covered by the WebFX kit (see the [JavaDoc](#) for coverage)
- be compatible with GWT and/or TeaVM (avoid reflection, multithreading, and blocking code)

When a JavaFX application meets these two requirements, we call it a *WebFX application*, and it can be transpiled for the web simply by running a GWT or TeaVM compilation of it together with the WebFX kit.

The main JavaFX features not currently covered are: JavaFX CSS, FXML, 3D, and most complex controls such as ComboBox, TableView, and TreeView. We plan to support CSS (using efficient browser CSS mapping rather than JavaFX CSS emulation) and provide simpler alternative WebFX controls to ComboBox and TableView before considering their actual emulation.



OpenJFX is a huge library (about 10MB) compared to standard JS frameworks (typically 100KB). Completing its coverage will take time, but you can already do most of what you can do with other standard frameworks.

Getting started

Prerequisites

To develop WebFX applications, you will need the following software already installed on your development machine:

- JDK 13 or above
- Maven
- Git
- Your preferred Java IDE

Introducing the WebFX CLI

The WebFX CLI is an essential Command Line Interface tool that will assist you in developing

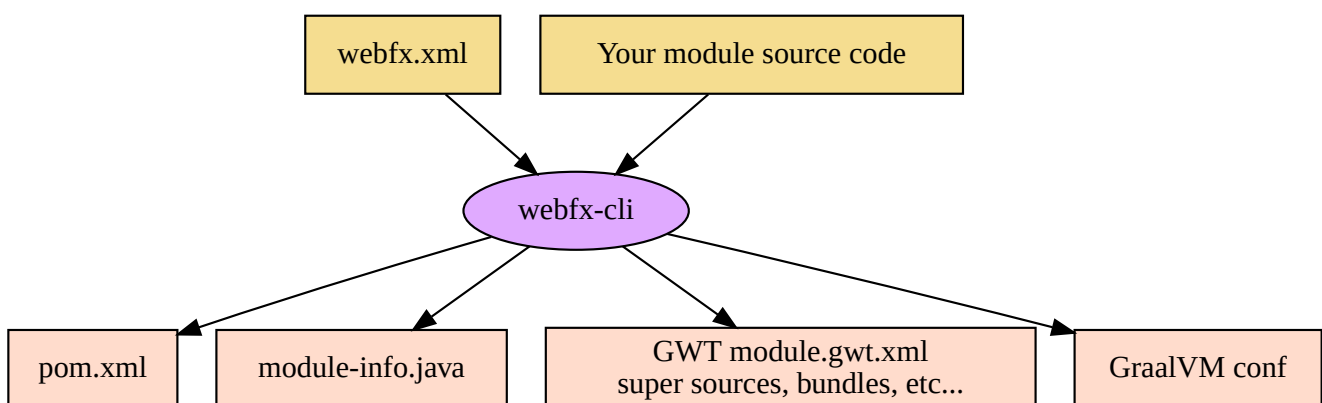
WebFX applications. It will create your application modules as follows:

Your repository

- └ xxx-application (1)
- └ xxx-application-gluon (2)
- └ xxx-application-gwt (3)
- └ xxx-application-openjfx (4)
- └ xxx-application-teavm-js (5)
- └ xxx-application-teavm-wasm (6)

- ① This module contains the JavaFX code of your application. It is cross-platform (not yet bound to a specific platform) and therefore not executable.
- ② This module targets the native desktop & mobile platforms. It binds your application with the OpenJFX runtime, and can call the Gluon toolchain to produce the Windows, macOS, Linux, Android & iOS native executables (depending on the OS of your local machine).
- ③ This module targets the web platform (JavaScript) via GWT. It binds your application with the WebFX kit and the GWT JavaScript compiler.
- ④ This module targets the standard desktop platform. It binds your application with the OpenJFX runtime, and is directly executable in your IDE. It can also call the standard Java toolchain to produce the desktop executables (Windows, macOS or Linux) with an embedded JRE.
- ⑤ This module targets the web platform (JavaScript) via TeaVM. It binds your application with the WebFX kit and the TeaVM JavaScript compiler.
- ⑥ This module targets the web platform (WebAssembly) via TeaVM. It binds your application with the WebFX kit and the TeaVM WebAssembly compiler.

You can create several WebFX applications in the same repository. As your application code grows, you can split your code into more modules. The CLI will help you to create and maintain all your modules. For each module, it will create and maintain your build chain as follows:



Your inputs will be centralized in the WebFX module files named `webfx.xml` (same location as `pom.xml`), and the CLI will generate the rest of the build chain from them. For example, a typical directive in `webfx.xml` will be:

```
<dependencies>
  <used-by-source-modules/>
</dependencies>
```

This directive asks the CLI to identify the list of your dependencies from an analysis of your source code, and automatically populate the dependencies in `pom.xml`, `module-info.java`, `module.gwt.xml`, etc...

During that process, the CLI takes care of the cross-platform aspects: when a feature is platform-dependent (a different implementation exists for different platforms), it will pick the right modules (those whose implementation matches the target platform). It is at this point, for example, that the CLI will replace the OpenJFX modules with the WebFX kit modules in your GWT and TeaVM application modules.

Installing the WebFX CLI

Since we haven't published an official WebFX release yet, the way to install the CLI for now is to clone the [webfx-cli](#) repository, and build it with Maven.



We will distribute the CLI in a better way with the first WebFX official release.

Cloning the webfx-cli repository

SSH

```
git clone git@github.com:webfx-project/webfx-cli.git
```

HTTPS

```
git clone https://github.com/webfx-project/webfx-cli.git
```

Building webfx-cli with Maven

This is achieved by running the Maven *package* goal under the `webfx-cli` directory:

```
cd webfx-cli
mvn package
```



As previously mentioned, WebFX CLI requires JDK 13 or above to successfully compile.

This generates an executable fat jar in the target folder that you can execute with java:

```
java -jar target/webfx-cli-0.1.0-SNAPSHOT-fat.jar
```

The `webfx.sh` and `webfx.bat` script files (located under the `webfx-cli` directory) are simply executing the same fat jar. We will use them in the next step.

Creating a permanent *webfx* alias

To easily invoke the CLI from a terminal, we need to create a permanent *webfx* alias. Run these commands in the `webfx-cli` directory:

Linux

```
echo "alias webfx='sh \"$(cd \"$(dirname \"$1\")\" && pwd -P)/$(basename \"$1\")/webfx.sh\"'" >> ~/.bashrc ①  
  
source ~/.bashrc ②
```

① Adding the alias to the shell profile

② Applying it to the current session

macOS >= Catalina

```
echo "alias webfx='sh \"$(cd \"$(dirname \"$1\")\" && pwd -P)/$(basename \"$1\")/webfx.sh\"'" >> ~/.zshrc ①  
  
source ~/.zshrc ②
```

① Adding the alias to the shell profile

② Applying it to the current session

macOS < Catalina

```
echo "alias webfx='sh \"$(cd \"$(dirname \"$1\")\" && pwd -P)/$(basename \"$1\")/webfx.sh\"'" >> ~/.bash_profile ①  
  
source ~/.bash_profile ②
```

① Adding the alias to the shell profile

② Applying it to the current session

Windows

You can simply add the `webfx-cli` repository to your environment path, so `webfx.bat` will be directly executed when typing the `webfx` command.

Now you should be able to invoke the CLI from the terminal:

```
webfx --help
```

Updating the WebFX CLI to the latest version

You can check for updates at any time by running:

```
webfx bump cli
```

If a new version is available, it will download it and build it for you.



This is so far the only command that uses `git` (a `git pull` of the `webfx-cli` repository). The CLI will not call `git` on your own repositories.

Creating your first WebFX app

Creating and initializing your repository

Let's create our first WebFX application. We need to create the repository directory and ask the CLI to initialize it, passing the `groupId`, `artifactId`, and `version` of our application.

```
mkdir webfx-example
cd webfx-example
webfx init org.example:webfx-example:1.0.0-SNAPSHOT
```

The `init` command creates only two files: `webfx.xml` and `pom.xml`. Note that if this is the first time you have used the CLI, it will download some other files through Maven to retrieve essential information about the available WebFX modules.

Creating your application modules

When you create an application, you pass the fully qualified name of the JavaFX class you want to create, and the prefix to use for the application modules:

```
webfx create application --prefix webfx-example
org.example.webfxexample.WebFxExampleApplication --helloWorld
```

This command created the following modules:

```
webfx-example
├─ webfx-example-application
├─ webfx-example-application-gluon
├─ webfx-example-application-gwt
├─ webfx-example-application-openjfx
├─ webfx-example-application-teavm-js
└─ webfx-example-application-teavm-wasm
```


The JavaFX class is located in the first module. Normally its `start()` method is empty at this stage, but because we specified the `--helloWorld` option, it has been populated with this simple template:

```
public class WebFxExampleApplication extends Application {

    @Override
    public void start(Stage primaryStage) {
        primaryStage.setScene(new Scene(new StackPane(new Text("Hello world!")), 800,
600));
        primaryStage.show();
    }

}
```

Building your application

The following command will do a simple build of your application (without generating any final executable):

```
webfx build
```

To generate the executables for the different platforms, you need to pass some extra build options. You can find the list in the build help:

```
webfx build --help
```

Some builds require the installation of third-party software, as shown in the following table:

Target platform	Build platform	Install command	Build option	Executable file(s)
Web	Linux, macOS or Windows		<code>--gwt</code>	HTML
Web	Linux, macOS or Windows		<code>--teavm</code>	HTML
Any desktop OS with Java	Linux, macOS or Windows		<code>--openjfx-fatjar</code>	fat jar
Linux (embedded JRE)	Linux	ubuntu-tools	<code>--openjfx-desktop</code>	executable, .AppImage, .rpm, .deb
macOS (embedded JRE)	macOS	xcode	<code>--openjfx-desktop</code>	executable, .dmg, .pkg
Windows (embedded JRE)	Windows	wix*, inno*	<code>--openjfx-desktop</code>	executable, .msi, .exe

Target platform	Build platform	Install command	Build option	Executable file(s)
Linux (native)	Linux	graalvm, ubuntu-tools	--gluon-desktop	executable
macOS (native)	macOS	graalvm, xcode-tools	--gluon-desktop	executable, .dmg, .pkg
Windows (native)	Windows	graalvm, vs-tools, wix*	--gluon-desktop	executable, .msi
Android (native)	Linux	graalvm, ubuntu-tools	--gluon-android (or --gluon-mobile)	.apk
iOS (native)	macOS	graalvm, xcode-tools	--gluon-ios (or --gluon-mobile)	.ipa

* these tools are optional, they are used to create installers (.msi or .exe)

We will now cover each target platform in detail.



You can open issues in the [webfx-cli](#) repository to report any bug you may encounter.

Web platform via GWT

The [GWT Maven plugin](#) will be invoked to generate the web application.

You can use the following commands:

Long syntax

```
webfx build --gwt ①
webfx build --gwt --locate ②
webfx build --gwt --show ③
webfx run --gwt ④
webfx build --gwt --run ⑤
```

- ① Build the HTML executable file with the GWT Maven plugin
- ② Locate the generated executable file
- ③ Show the generated executable file in the file explorer
- ④ Execute the generated executable in the browser
- ⑤ Build the app and execute it in a single command

Short syntax

```
webfx build -g ①
webfx build -gl ②
webfx build -gs ③
webfx run -g ④
```

```
webfx build -gr ⑤
```

- ① Build the HTML executable file with the GWT Maven plugin
- ② Locate the generated executable file
- ③ Show the generated executable file in the file explorer
- ④ Execute the generated executable in the browser
- ⑤ Build the app and execute it in a single command



As opposed to `--show`, `--locate` works even before the build, as it prints the expected location, whether the executable file is present or not.

The `run` command has exactly the same options as the `build` command. The only difference between these two commands is that with `run` you can run the app without rebuilding it (unless `--build` is specified), and with `build` you can build the app without running it (unless `--run` is specified).

Web platform via TeaVM

The [TeaVM Maven plugin](#) will be invoked to generate the web application.

You can use the following commands:

Long syntax

```
webfx build --teavm ①  
webfx build --teavm --locate ②  
webfx build --teavm --show ③  
webfx run --teavm --javascript ④  
webfx build --teavm --javascript --run ⑤
```

- ① Build the HTML executable files (JavaScript and WebAssembly) with the TeaVM Maven plugin
- ② Locate the generated executable files
- ③ Show the generated executable files in the file explorer
- ④ Execute the generated JavaScript executable in the browser
- ⑤ Build the JavaScript app and execute it in a single command

Short syntax

```
webfx build -t ①  
webfx build -tl ②  
webfx build -ts ③  
webfx run -tj ④  
webfx build -tjr ⑤
```

- ① Build the HTML executable files (JavaScript and WebAssembly) with the TeaVM Maven plugin
- ② Locate the generated executable files

- ③ Show the generated executable files in the file explorer
- ④ Execute the generated JavaScript executable in the browser
- ⑤ Build the JavaScript app and execute it in a single command



`webfx run --teavm --wasm` (short syntax `webfx run -tw`) does not work because `http(s)` is required for running WebAssembly. But you can use the built-in web server of your Java IDE to serve the file located by `webfx build -twl`.

In the next sections, we won't repeat all these commands; we will use only the last command to build and run the application in one go, but you can still choose to separate build and run.

Desktop platform (JAR)



The desktop JAR executable is usually used for local testing. If you want to distribute your desktop application, the embedded JRE or native executables are preferred, as they don't require Java installed on the users' machines.

The [Maven Shade plugin](#) will be invoked to generate an executable fat jar.

You can use the following command:

Long syntax

```
webfx build --openjfx-fatjar --run
```

Short syntax

```
webfx build -fr
```



You can combine the build options. For example `webfx build --gwt --openjfx -fatjar` (short syntax: `webfx build -gf`) will build both the GWT HTML and OpenJFX fat jar executables.

Desktop platform (embedded JRE)

As opposed to the JAR, the embedded JRE will be specific to the target platform. Therefore, a Linux machine will build a Linux executable, a Mac will build a macOS executable, and a Windows machine will build a Windows executable.

The [JavaPackager Maven plugin](#) will be invoked to generate the executables and installers for the different platforms.

Prerequisite: you must install the following software for a successful build:

Linux

```
webfx install ubuntu-tools ①
```

① These tools are required by JavaPackager.

macOS

```
webfx install xcode ①
```

① Just checks that Xcode is installed with the correct path, otherwise suggests solutions.

Windows

```
webfx install wix ①  
webfx install inno ②
```

① Optional. The WiX Toolset is used by JavaPackager to create a simple .msi installer.

② Optional. Inno Setup is used by JavaPackager to create a more elaborate .exe installer.



The CLI will help you to customize the installers (application name, icon, etc...) in a future version. For now, we just use the default settings.



On macOS, you need to configure at least a Team ID in Xcode to make the application runnable on your machine. We will provide a guide for this later.

Then, you can use the following command:

Long syntax

```
webfx build --openjfx-desktop --run
```

Short syntax

```
webfx build -kr
```



If the application fails to run on macOS, you can try `webfx run -k --open` to execute it via the `open` command. This can open a problem report with more information.

Desktop platform (native)

As with the embedded JRE, three different machines are required to target the Linux, macOS, and Windows executables.

The [GluonFX Maven plugin](#) will be invoked to generate the native desktops.

Prerequisites: you must install the following software for a successful build:

Linux

```
webfx install graalvm ①  
webfx install ubuntu-tools ①
```

① Required for the GluonFX Maven plugin

macOS

```
webfx install graalvm ①  
webfx install xcode ②
```

① Required for the GluonFX Maven plugin

② Just checks that Xcode is installed with the correct path, otherwise suggests solutions.

Windows

```
webfx install graalvm ①  
webfx install vs-tools ①  
webfx install wix ②
```

① Required for the GluonFX Maven plugin

② Optional. The WiX Toolset is called by the GluonFX Maven plugin to create a simple .msi installer.



On macOS, you need to configure at least a Team ID in Xcode in order to make the application runnable on your machine. We will provide a guide for this later.

Then, you can use the following command:

Long syntax

```
webfx build --gluon-desktop --run
```

Short syntax

```
webfx build -dr
```



On machines that don't support OpenGL (such as some Linux virtual machines), JavaFX may report: Error initializing QuantumRenderer: no suitable pipeline found. We plan to provide an option later to include the JavaFX software pipeline as a backup, but this will increase the size of the executable.

Android platform

A Linux machine is required to build the Android executable.

The [GluonFX Maven plugin](#) will be invoked to generate the native Android application.

Prerequisites: you must install the following software for a successful build:

```
webfx install graalvm
webfx install ubuntu-tools
```

Then, you can use the following command:

Long syntax

```
webfx build --gluon-android --run
```

Short syntax

```
webfx build -ar
```

Alternatively, you can use the `--gluon-mobile` (short: `-m`) cross-platform option. It will be interpreted as `--gluon-android` on Linux machines.

In this context, running the application means invoking the GluonFX Maven plugin to install and run the generated package on your Android device connected to your Linux machine via USB. For this to succeed, you must first activate the Developer mode and then the USB debugging option on your Android device.



Only 64-bit architecture is supported (the installation will fail on 32-bit devices). Please check that your Android kernel version specifies `x64`.

iOS platform

A macOS machine is required to build the iOS executable.

The [GluonFX Maven plugin](#) will be invoked to generate the native iOS application.

Prerequisite: you must install the following software for a successful build:

```
webfx install graalvm
webfx install xcode
```



You also need a provisioning profile that links the app to your iOS device, otherwise you won't be able to install it on your device. We will provide a guide for this later.

Then, you can use the following command:

Long syntax

```
webfx build --gluon-ios --run
```

Short syntax

```
webfx build -ir
```

Alternatively, you can use the `--gluon-mobile` (short: `-m`) cross-platform option. It will be interpreted as `--gluon-ios` on Macs.

GitHub workflows

In a future release of the CLI we plan to support automatic generation of GitHub workflows. Each push to the main branch of your GitHub repository will then trigger the generation of all your executables on the Linux, macOS and Windows GitHub servers.

For now, you can check out the demos to see sample GitHub workflow configuration. For example, the [FX2048 GitHub workflow](#) and the [generated executables](#) (expand the [Assets](#) link to view them).

Developing in your IDE

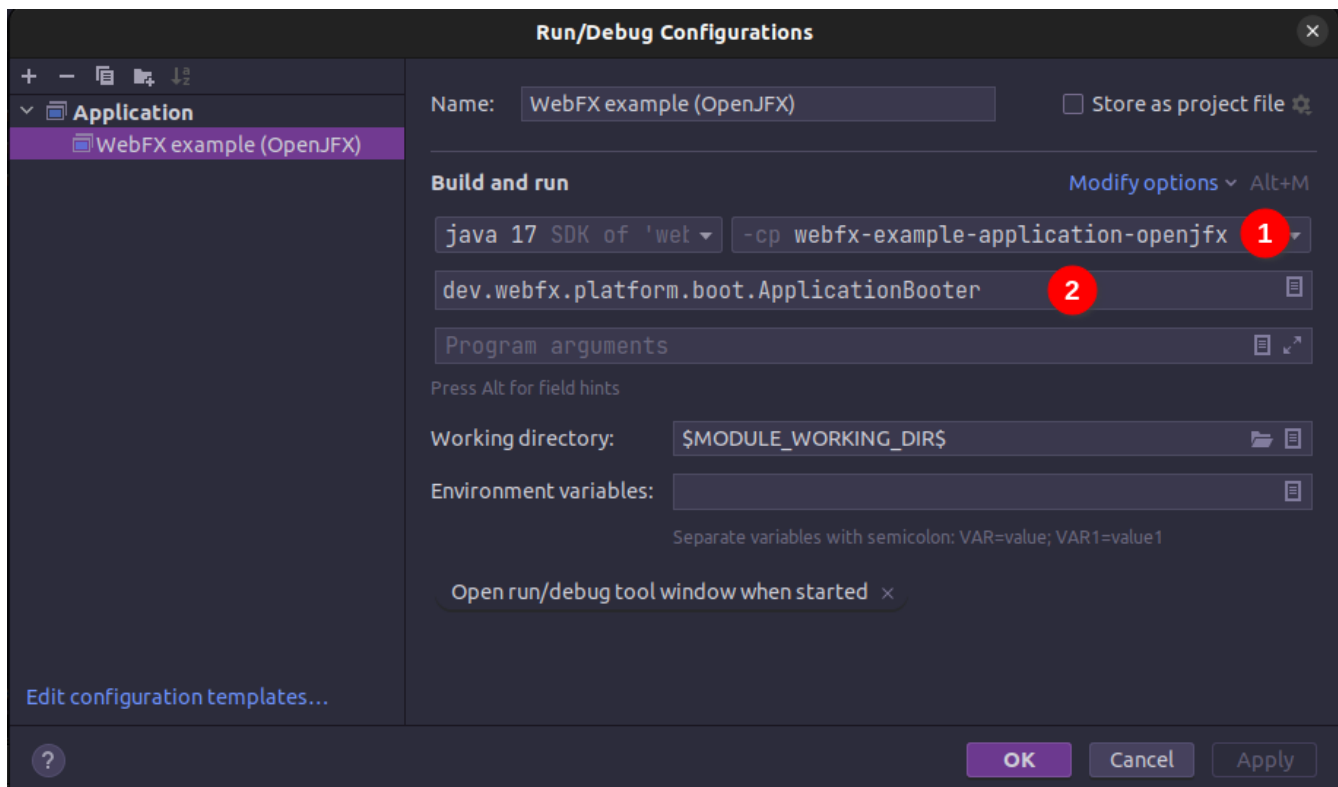
Instructions for IntelliJ IDEA are provided below, which can be easily adapted to other Java IDEs.

Opening the project

Open the `webfx-example` directory from your Java IDE. It should recognize the directory as a Maven project, and import it.

Building and running the OpenJFX application

Create an application configuration as follows:



- ① Select the OpenJFX application module
- ② Enter `dev.webfx.platform.boot.ApplicationBooter` for the main class



You can just type `AB` for the main class, and your IDE should quickly find and suggest the WebFX ApplicationBooter class.

The way to boot GWT, TeaVM and OpenJFX applications is different, but WebFX offers a cross-platform way to do it. For this reason, the main class of a WebFX application is always `dev.webfx.platform.boot.ApplicationBooter`. It will find your JavaFX application because it has been automatically declared as a Java service by the CLI.

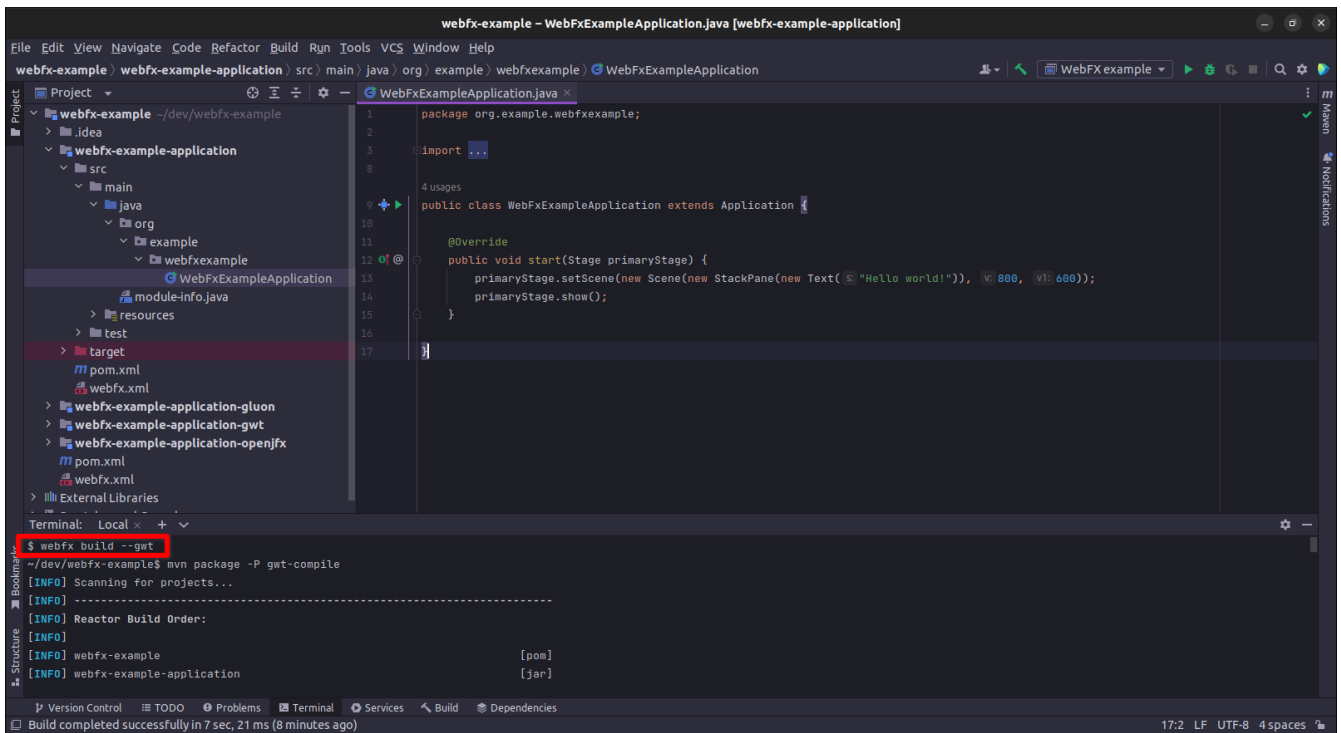


GWT normally doesn't support the Java service API, but WebFX does, because the CLI emulates it by generating a GWT super source. You can rely on this feature to declare and implement your own services. Your services can even have platform-dependent implementations. For example, a service can be a cross-platform UI API, with an OpenJFX implementation and a different GWT/TeaVM implementation using the JS library you want for your web app. We will document this later.

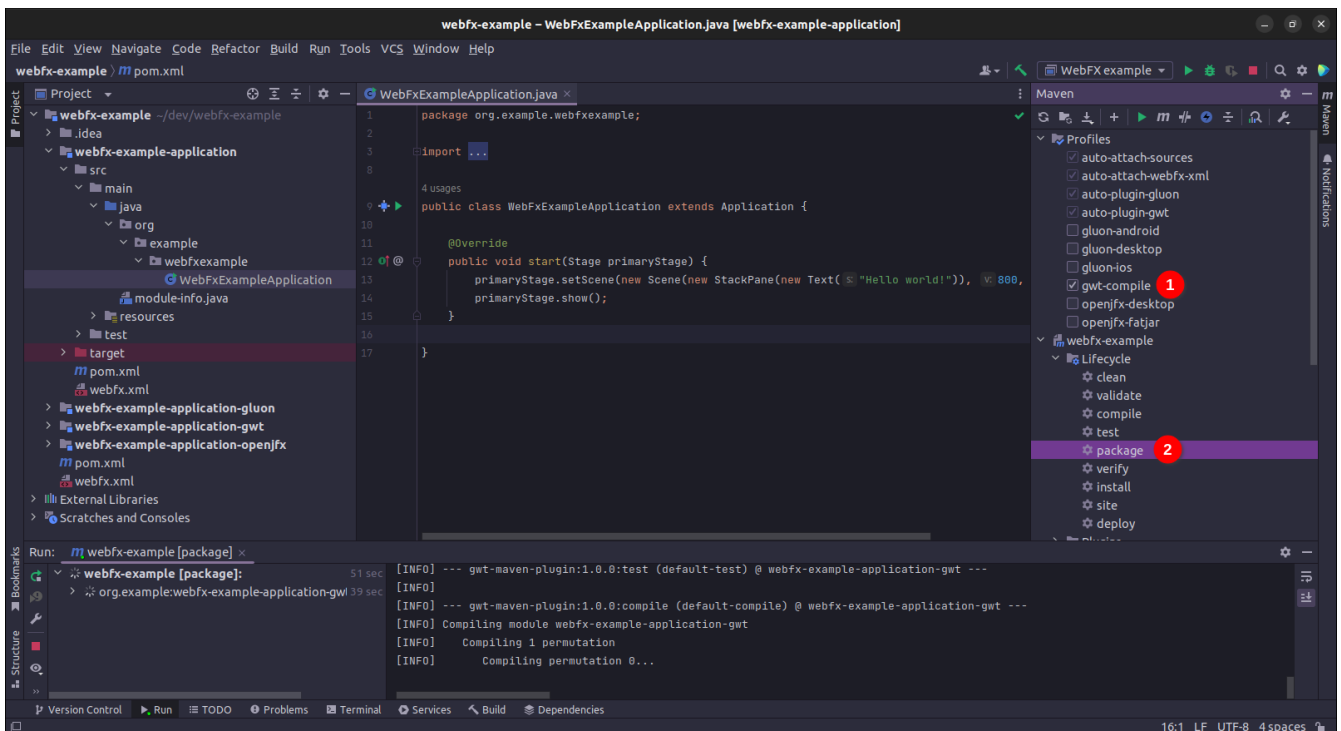
If you run this configuration, it will build and run your WebFX application in your IDE using the OpenJFX runtime. This is the configuration that you will use to develop and debug your application.

Building and running the GWT application

Unlike the OpenJFX version where the IDE can use its own Java build system, the GWT version can only be built through Maven. One way to build it is to open a terminal window in your IDE, and to type `webfx build --gwt`.

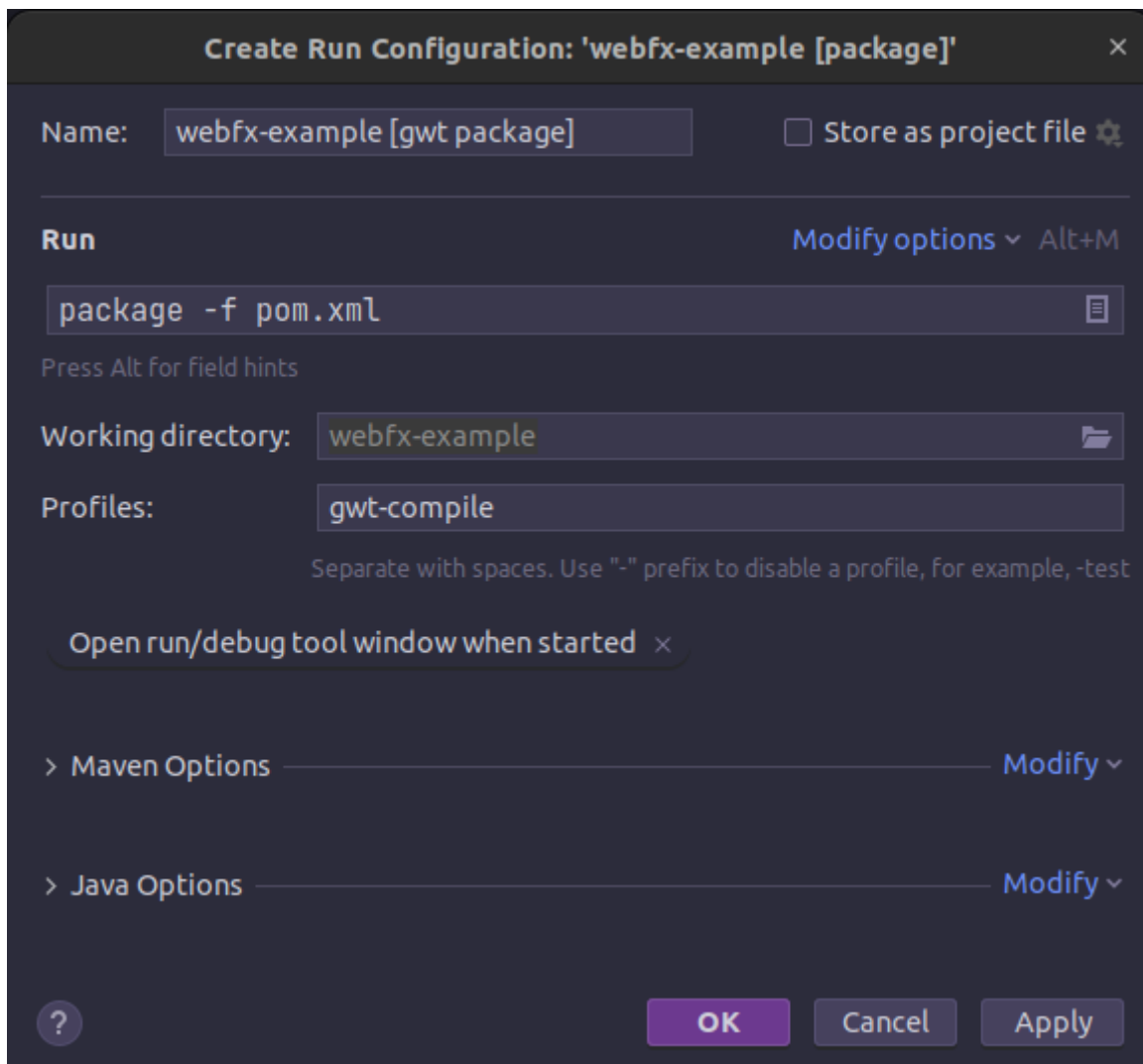


Another way is to trigger the build manually through the IDE Maven window.

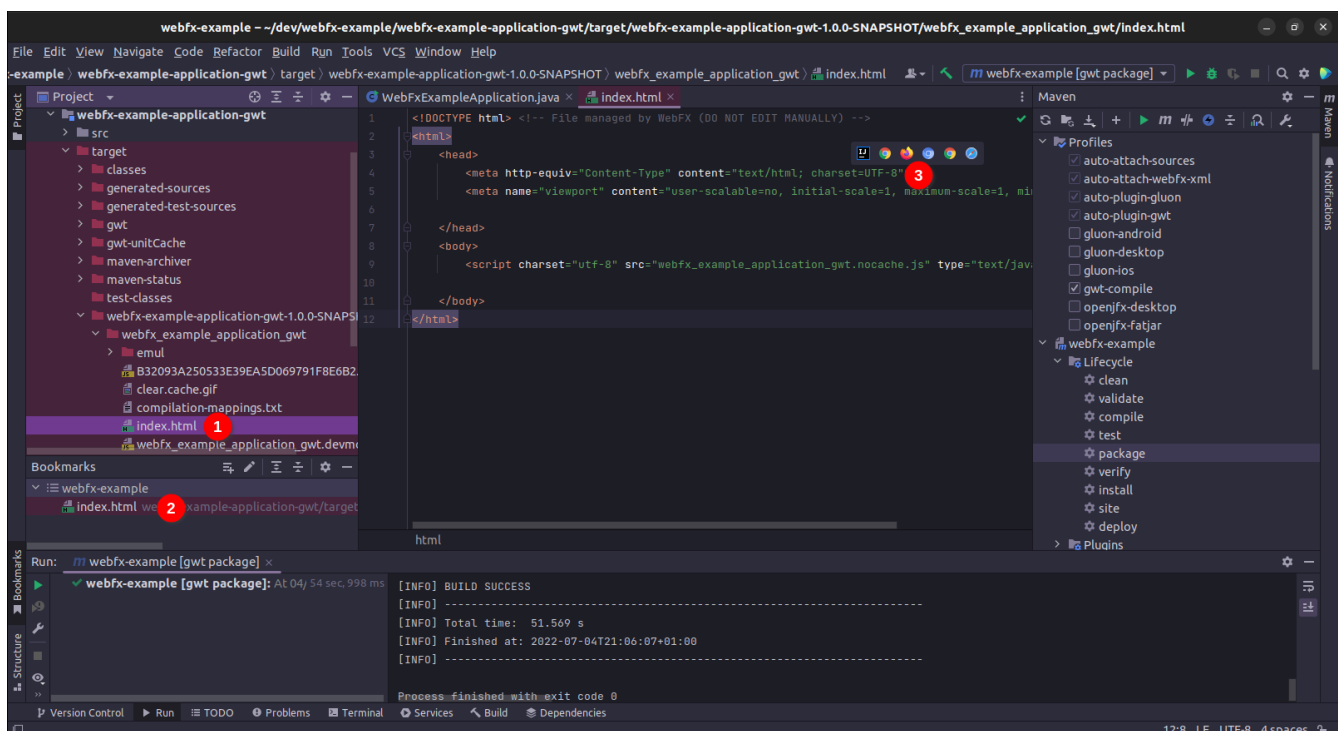


- ① Activate the **gwt-compile** Maven profile
- ② Run the **package** Maven goal

It can be a good idea to create a configuration by selecting **Modify Run Configuration** in the context menu of that **package** goal, and press OK in the dialog. You can then easily run this configuration to trigger the GWT build.



Once built, you can ask your IDE to open the generated HTML file in a browser.



- ① Locate the generated HTML file

- ② You can drag & drop it in the Bookmarks window for later
- ③ Select the browser you want to use to open it

Making changes

Let's do a first small change by replacing the `Text` with a `Button`. At first sight, it looks like an insignificant change, but actually it's not, because this involves a new dependency on the `javafx-controls` module (our application was only using the `javafx-graphics` module so far). To take advantage of the CLI, let's enter the fully qualified `Button` class name like this:

```
public class ExampleApplication extends Application {  
  
    @Override  
    public void start(Stage primaryStage) {  
        primaryStage.setScene(new Scene(new StackPane(new  
javafx.scene.control.Button("Hello world!")), 800, 600));  
        primaryStage.show();  
    }  
  
}
```

The IDE doesn't recognize this class yet, but let's ask the CLI to update the build chain from the terminal window:

```
webfx update
```

The CLI should make all necessary changes in the `pom.xml`, `module-info.java`, and `module.gwt.xml` files. After a few seconds, your IDE should automatically detect and apply these changes. Then it should recognize the `Button` class, and allow you to import it.

Finally, let's add a simple code to react to the button:

```
public class ExampleApplication extends Application {  
  
    @Override  
    public void start(Stage primaryStage) {  
        Button button = new Button("Hello world!");  
        button.setOnAction(event -> button.setText("You clicked me!"));  
        primaryStage.setScene(new Scene(new StackPane(button), 800, 600));  
        primaryStage.show();  
    }  
  
}
```

First, check that the code works as expected with the OpenJFX runtime by running the OpenJFX configuration. Then build and run the GWT/TeaVM version using one of the methods already

explained.

Our application looks a bit sad? Maybe you can try [this little code](#) that adds a bit of color, animation, and effects?

Got more creative ideas? Now is the time to play with WebFX!